MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, FLUOROELASTOMER, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

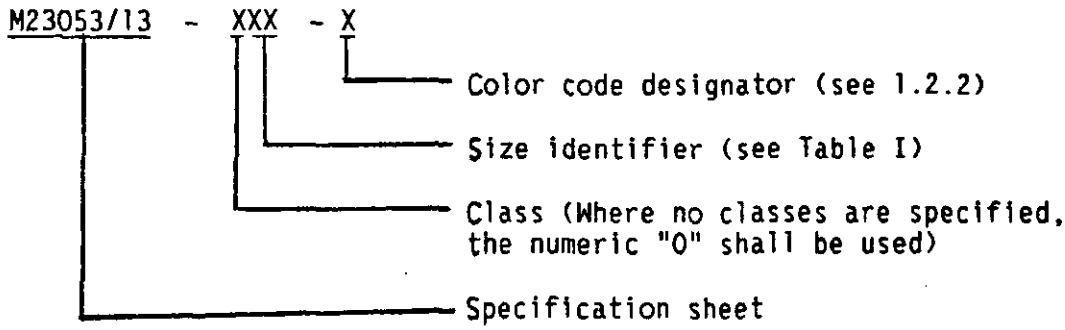
REQUIREMENTS:

Continuous operating temperature range: -40°C (-40°F) to +200°C (+392°F).

Color: The sleeving shall be furnished in a black color that conforms to the requirements of Class II, MIL-STD-104.

Longitudinal change: ± 20 percent.

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



Example: Black, 0.500 inch (12.7 mm) as supplied ID sleeving shall be identified as M23053/13-004-0.

NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970

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TABLE I. Construction details, inches (mm). 1/

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
M23053/13-001-0	.187(4.7)	.094(2.4)	.035 ± .007(.9 ± .18)
M23053/13-002-0	.250(6.4)	.125(3.2)	.035 ± .007(.9 ± .18)
M23053/13-003-0	.375(9.5)	.187(4.7)	.035 ± .007(.9 ± .18)
M23053/13-004-0	.500(12.7)	.250(6.4)	.035 ± .007(.9 ± .18)
M23053/13-005-0	.625(15.9)	.312(7.9)	.042 ± .008(1.1 ± .20)
M23053/13-006-0	.750(19.1)	.375(9.5)	.042 ± .008(1.1 ± .20)
M23053/13-007-0	.875(22.2)	.437(11.1)	.049 ± .011(1.2 ± .28)
M23053/13-008-0	1.000(25.4)	.500(12.7)	.049 ± .011(1.2 ± .28)
M23053/13-009-0	1.250(31.8)	.625(15.9)	.055 ± .015(1.4 ± .38)
M23053/13-010-0	1.500(38.1)	.750(19.1)	.055 ± .015(1.4 ± .38)
M23053/13-011-0	2.000(56.8)	1.000(25.4)	.065 ± .017(1.7 ± .43)

1/ Diameter limits for object to be enclosed shall be as specified in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ Black is the standard color.

Unrestricted shrinkage: Test method 4.6.5; 200° ± 5°C (392° ± 9°F) for 10 minutes.

TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Cold impact	No cracking	4.6.7.2 -40° ± 1°C (-40° ± 2°F)
Heat shock	No cracking, flowing or dripping	4.6.8 300° ± 5°C (572° ± 9°F)
Restricted shrinkage	No cracks	4.6.6.1.2 200° ± 2°C (392° ± 4°F)
Voltage withstand	Pass	4.6.6.2

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile stength, psi (MPa), minimum	1,200 (8.2)	4.6.14, ASTM D 412
Ultimate elongation, percent, minimum	250	4.6.14, ASTM D 412
Tensile stress at 200 percent elongation, psi (MPa), maximum	2,000(13.8)	4.6.13, ASTM D 412
Dielectric strength, volts/mil (Kv/mm), minimum	200(7.9)	4.6.2 ASTM D 2671
Volume resistivity, Ohm-cm minimum	10^9	4.6.2 ASTM D 876
Corrosion	No corrosion or pitting	4.6.10.1 and 4.6.10.2 175° ± 2°C (347° ± 4°F) for 16 hours.
Heat resistance, properties after:		4.6.9 250° ± 3°C (482° ± 5.4°F), for 168 hours
Tensile strength, psi (MPa), miminum	1,200(8.2)	
Ultimate elongation, percent, minimum	200	

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Water absorption at 23°C, percent, maximum	0.5	4.6.2 ASTM D 570, 24 hrs
Fluid resistance, properties after:		4.6.11
Tensile strength, psi (MPa), minimum	1,200 (8.2)	
Ultimate elongation, percent, minimum	250	
Flammability	Self-extinguishing within 15 seconds	4.6.15.2
Fungus resistance	No growth	ASTM G 21

Shelf life: The supplier shall certify storage at 40°C (104°F) for a period of two years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: Heat shrinkable, flexible, fluoroelastomer sleeving is used in high temperature applications or where exposure to high temperature solvents is expected.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - MI
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

(Project 5970-0577-12)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23053 /13A	INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABL GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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